

Pablo Corbalán

p.corbalanpelegrin@unitn.it | +39 391 774 8082 | pablocorbalan.com

Contact

Via Sommarive 9
38123 Trento, Italy

 pablocorbalan

 pablocorbalan

 paulcorbalan

Languages

Spanish (native)
English (fluent)
Catalonian (basic)

Programming

Python | C
JavaScript | HTML
Java | C++
MATLAB | node.js

Software

Matplotlib | Pandas
NumPy | SciPy
Contiki OS | Cooja
Eagle | Proteus
Adobe InDesign
LaTeX | Microsoft Office
Git | GitHub | GitLab | SVN

RF Localization

UWB SS-TWR | DS-TWR
UWB TDOA
UWB Concurrent Ranging
BLE RSSI

HW Platforms

DecaWave EVB1000
DecaWave DWM1001
TI CC2650 LaunchPad
TelosB | CC2538 OpenMote
Raspberry Pi
BeagleBone Black
nRF52832 | STM32F105

Protocols

RPL | ORPL | CTP
ContikiMAC | CSMA | TSCH
RIME | 6LoWPAN
UDP | TCP
CoAP | HTTP

Other

April 2012 - March 2013
**IEEE UMH Student Branch
Chairman & Co-founder**

Oct. 2010 - Oct. 2012
**UMH Student
Representative**

Education

September 2016 - Present (expected April, 2020)

PhD Candidate

University of Trento, Italy

Energy-efficient and Pervasive Ultra-wideband Localization.

October 2014 - August 2016

PhD Candidate

Nimbus Centre, Cork Institute of Technology, Ireland

Network-wide Management and Control in Wireless Sensor Networks (WSNs). **Resigned.**

September 2009 - October 2014

Telecommunication Engineering

Miguel Hernández University, Spain

Special Mentions in Circuit Theory 1 & 2, Electronic Circuits, Printed Circuits, Digital Electronic Systems, Photonics, and Advanced Programming. **Grade: 8.26 / 10.**

September 2013 - June 2014

Electronic Engineering

Cork Institute of Technology, Ireland

MSc Thesis at the Nimbus Centre on WSN Application Level Protocols. **Erasmus Exchange.**

Work Experience

May 2019 - Present

Wireless Systems Intern

Robert Bosch LLC, Sunnyvale, California, USA

Research on ultra-wideband ranging schemes and localization algorithms.

September 2017 - Present

Teaching Assistant & Supervision

University of Trento, Italy

Laboratory of Wireless Sensor Networks. MSc in Computer Science and Telecommunication.

3 MSc Theses Co-supervised with G. P. Picco on UWB Communication and Localization.

October 2014 - December 2014

Part-time Developer

Nimbus Centre, Cork Institute of Technology

Maintenance and update of a 30-node WSN deployment in a university data centre.

July 2014 - September 2014

Erasmus Internship

Nimbus Centre, Cork Institute of Technology

Development of drivers, Contiki applications, and a gateway for two WSN deployments.

March 2013 - August 2013

R&D Internship

emxys

PCB design and testing for space and commercial applications. Funded by Santander Bank.

Selected Publications

V. Magnago, **P. Corbalán**, G. P. Picco, L. Palopoli, and D. Fontanelli. **Robot Localization via Odometry-assisted Ultra-wideband Ranging with Stochastic Guarantees.** IROS, Nov. 2019.

D. Vecchia, **P. Corbalán**, T. Istomin, and G. P. Picco. **TALLA: Large-scale TDoA Localization with Ultra-wideband Radios.** In Proc. of IPIN, October 2019. **Best Paper Award.**

D. Vecchia, **P. Corbalán**, T. Istomin, and G. P. Picco. **Playing with Fire: Exploring Concurrent Transmissions in Ultra-wideband Radios.** In Proc. of IEEE SECON, June 2019.

P. Corbalán, G. P. Picco, and S. Palipana. **Chorus: UWB Concurrent Transmissions for GPS-like Passive Localization of Countless Targets.** In Proc. of IEEE/ACM IPSN, April 2019.

P. Corbalán and G. P. Picco. **Concurrent Ranging in Ultra-wideband Radios: Experimental Evidence, Challenges, and Opportunities.** In Proc. of EWSN, Feb. 2018. **Best Paper Award.**

R. Marfievici, **P. Corbalán** et al. **Small Tales from the C130 Horror Room: A Wireless Sensor Network Story in a Data Center.** In Proc. of FAILSAFE, November 2017.

Awards

Best Paper Award

IPIN '19

Best Paper Award

EWSN '18

Best Poster Award

EWSN '16

Five Star University Student

Miguel Hernández University, April 2016

Best Telecommunication Student

COITCV, March 2015